MIL-C-45359A
20 December 1967
SUPERSEDING
MIL-C-45359
9 March 1961

MILITARY SPECIFICATION

COIL, STARTING: ENGINE IGNITION, WITH FILTER,
24 VOLT DC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers waterproof, 24 volt (nominal rating) direct current (DC) starting coils for use on ignition systems incorporating direct coupled magnetos used on internal combustion engines for military vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Federal
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
Military
MIL-C-11693 - Capacitors, Feed Through, Radio-Interference Reduction, AC and DC (Hermetically Sealed in Metallic Cases) General Specification For.
MIL-E-13856 - Electrical Components for Automotive Vehicles; Waterproofness and Resistance to Temperature Cycling Tests, General Requirements for.
MIL-F-13927 - Fungus Resistance Test; Automotive Components.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARDS
Federal
FED. STD. NO. 595 - Colors.
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking for US Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-202 - Test Methods for Electronic and Electric Component Parts.
MS 51355 - Coil, Starting, Engine Ignition, with Filter, 24 Volt DC.

FSC-2920

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DRAWINGS
Army
C7727531 - Gap, Sparking.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Qualification. The coils furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Design and construction. The coils shall conform to MS 51355, or other applicable standards or drawings as specified (see 6.2).

3.2.1 Interchangeability. To provide for interchangeability, coils fabricated in accordance with this specification shall have external dimensions that fall within the envelope outline of the applicable drawing or military standard. Electrical connectors, mounting dimensions and location shall also conform to the detail requirements of the applicable document.

3.2.2 Materials. Materials shall be as specified herein, on applicable standards or drawings, and in applicable specifications.

3.2.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided. Separation of dissimilar metals shall be accomplished by providing insulation between mating surfaces.

3.2.3 Standard parts. Military standard parts shall be incorporated wherever applicable. Commercial standard parts may be used provided they are interchangeable with military standard parts without modification, and the equivalent military standard part is identified by the manufacturer.

3.2.4 Weight. The weight of the coil shall not exceed 30 ounces.

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3.2.5 Threaded parts. All screw threads shall conform to National Bureau of Standards Handbook H28. The form, number per inch, and class of threads shall be as specified on applicable standards or drawings (see 2.2).

3.2.6. Locking devices. Lock washers, self-locking nuts, safety wires, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components.

3.2.7 Solder. Solder employed in making electrical connections shall be a rosin core type conforming to QQ-S-571.

3.2.8 Capacitors. If used, feed through capacitors for radio interference suppression shall be in accordance with the applicable requirements of MIL-C-11693.

3.2.9 Polarity. The coil shall operate with a negatively grounded circuit.

3.3 Electrical.

3.3.1 Output voltage. When tested as specified in 4.6.1, the output voltage shall be not less than the values specified in table I.

Table I Minimum output voltage (Kilovolts (KV))

Input	Magneto Shaft Speed	Plus 77° ± 15°F Minus 65° ± 5°F	Plus 160° ± 10° F
Voltage	RPM	KV	KV
18	35	9.0	8.0
18	100	9.0	8.0
28.5	35	10.0	9.0
28.5	100	10.0	9.0

3.3.2 Endurance. When tested as specified in 4.6.2, the coil shall evidence no damage or malfunction and shall subsequently meet the requirements of 3.3.1.

3.3.3 Dielectric strength. When tested as specified in 4.6.3, the coil shall evidence no burning, charring, smoking, rupture, or other damage to the insulation and shall subsequently meet the requirements of 3.3.1.

3.3.4 Shunting resistance. When tested as specified in 4.6.4, the coil in conjunction with an associated magneto shall maintain an output voltage of not less than 5,000 volts when operating at a magneto shaft speed of 100 rpm, with an input of 28.5 volts DC, and a resistance of 250,000 ohms in a circuit having a total capacitance load of 150 ± 10 micromicrofarads.

3.4 Environmental.

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3.4.1 Vibration. When tested as specified in 4.7.1, the coil shall evidence no breakage, loosened or distorted parts or other physical damage and shall subsequently meet the requirements of 3.3.1.

3.4.2 Waterproofness. When tested as specified in 4.7.2, the coil shall evidence no leakage or damage which would impair electrical performance.

3.4.3 Shock. When tested as specified in 4.7.3, the coil shall evidence no breakage, loosened or distorted parts or other physical damage and shall subsequently meet the requirements of 3.3.1.

3.4.4 Fungus resistance. When tested as specified in 4.7.4, the coil shall evidence no microbial growth adversely affecting performance and shall subsequently meet the requirements of 3.3.3 and 3.3.1.

3.4.5 Corrosion resistance. When tested as specified in 4.7.5, the coil shall evidence no corrosion affecting performance and shall subsequently meet the requirements of 3.3.3 and 3.3.1.

3.5 Finish. Unless otherwise specified (see 6.2), surface treatment and painting of the housing and cover shall be in accordance with MIL-STD-193. Color shall be black gloss enamel conforming to color chip 17038 of FED. STD. NO. 595.

3.6 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as a minimum shall include the following:

Coil, Starting 24 Volt DC
Current Rating
Military Part Number
Federal Stock Number
Manufacturer's Identification
US

3.7 Workmanship. Workmanship shall be such as to assure a product free of rust, burrs, scratches, chips, sharp edges, loose or defective connectors, cracked insulation or faulty soldering.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification inspection. For the purpose of qualification (see 3.1) the prospective supplier shall submit four coils. These specimens, properly marked with identifying information, shall be representative of the units proposed to be furnished to the Government in connection with future procurement.

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Qualification testing shall be conducted by the Government at a place designated by the Government and shall consist of examination for the defects specified in 4.3.2, and testing as specified in table II.

Table II. Order of qualification tests

Specimen No.	Paragraph No.	Test Sequence
1	4.6.1	Output voltage
	4.6.2	Endurance
	4.6.1	Output voltage
	4.6.4	Shunting resistance
2	4.6.1	Output voltage (77° ± 15°F)
	4.7.1	Vibration
	4.7.3	Shock
	4.6.1	Output voltage (77° ± 15°F)
	4.7.5	Corrosion resistance
	4.6.3	Dielectric strength
	4.6.1	Output voltage (77° ± 15°F)
3	4.6.1	Output voltage (77° ± 15°F)
	4.7.4	Fungus resistance
	4.6.3	Dielectric strength
	4.6.1	Output voltage (77° ± 15°F)
4	4.6.1	Output voltage (77° ± 15°F)
	4.6.4	Shunting resistance
	4.7.2	Waterproofness
	4.6.3	Dielectric strength
	4.6.1	Output voltage (77° ± 15°F)

4.2.1 Failure. Failure of any specimen to pass any of the examinations or tests specified herein may be cause, at the option of the Government, for refusal to conduct additional testing until the faults revealed by the tests have been corrected.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 Lot formation. A lot shall consist of all coils of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

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4.3.1.3 Sampling for testing. Samples for quality conformance testing shall be selected in accordance with level S-3 of MIL-STD-105 from each lot that has passed the examination specified in 4.3.2.2.

4.3.2 Examination.

4.3.2.1 Acceptable quality level. Each coil selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table III.

Table III. Classification of defects

Categories	Defects
Major:	
101	Nonconformance to dimensions which affect interchangeability (see 3.2.1)
Minor:	
201	Nonconformance to dimensions which do not affect interchangeability (see 3.2.1)
202	Improper finish (see 3.5)
203	Improper marking (see 3.6)
204	Faulty workmanship (see 3.7)

4.3.3 Tests.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in table IV using an AQL of 6.5 on the basis of percent defective.

Table IV. Order of quality conformance testing

Paragraph	Test
4.6.1	Output voltage
4.6.3	Dielectric strength

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4.4 Control tests.

4.4.1 Sampling for control tests. Coils shall be selected at the rate of two of each 500 produced, provided that not more than four nor less than two coils shall be selected in any 30 day period. The control test sample shall be selected from a lot which has passed the examination specified in 4.3.2, and the tests specified in table IV, and shall be subjected to the tests listed in table V in the order listed.

Table V. Order of control testing

Paragraph	Test
4.6.4	Shunting resistance
4.6.3	Dielectric strength

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test, may be cause, at the option of the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the examinations or tests have been corrected.

4.5 Test procedures.

4.5.1 Conditions. Tests shall be conducted under general laboratory conditions which are free of dust, drafts and funes. The ambient room temperature of 77° ± 15°F shall be maintained. Mounting shall be in the manner and with devices for securing the coil as in intended use. Unless otherwise specified herein or in applicable specifications, test specimens shall be thermally stabilized for one hour before being subjected to the test.

4.5.2 Apparatus. In addition to the standard measuring equipment, (see table VI and applicable test specification), the apparatus shall include a testing chamber capable of maintaining specified air temperatures over the range from minus 65°F to plus 165°F. Air movement in the high temperature chamber shall be held to a minumum to prevent a temperature gradient within the chamber and any cooling resulting from excess air motion.

4.5.3 Calibration of test equipment. Unless specified herein, test equipment accuracy shall be such as to permit measurement of 10 percent of product or test specification tolerance. Calibration of test equipment shall be conducted at intervals sufficient to establish required accuracy. Records of calibration shall be made available to the Government. The inspector may refuse to conduct an inspection where accuracy of test equipment has not been established to the satisfaction of the Government. Required measuring instrument tolerances are as listed in table VI.

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Table VI. Measuring instrument tolerance

Instrument	Accuracy - Maximum
Voltmeter DC	± 0.25 percent
Electrostatic voltmeter	± 1.0 percent
Oscilloscope	± 3.0 percent

4.6 Electrical.

4.6.1 Output voltage. To determine conformance to 3.3.1, the coil shall be conditioned for four hours at an ambient air temperature of 77° ± 15°F. The coil shall be energized with an input voltage of 18.0 ± 0.5 volts DC and coil terminal (booster and filter circuit terminal) shall be connected to a magneto with which the coil is intended to be used. One magneto secondary lead having voltage compatible with the polarity of the coil shall be connected through a type 1B3GT rectifier to a capacitor of 0.001 microfarads. The voltage across the capacitor shall be measured with an electrostatic voltmeter. Alternate suitable voltage measuring methods may be substituted. Total capacitance of the secondary circuit including the measuring apparatus shall be 150 ± 10 micromicrofarads. The other secondary leads shall be connected to standard sparking gap Drawing C7727531 set to an opening of three millimeters. Connections shall be made with suitable high-tension cable. The total capacitance of each such secondary circuit to ground including leads and any external capacitor shall be 150 ± 10 micromicrofarads. The magneto shall be operated at 35 and 100 rpm and the output voltage at each speed recorded. Subsequently the test shall be repeated at an input voltage of 28.5 ± 0.5 volts DC. The tests shall be repeated except that the ambient air temperature shall be plus 160° ± 10 F and minus 65° ± 5°F.

4.6.2 Endurance. To determine conformance to 3.3.2, the coil with input lead connected to a 28.5 ± 0.5 volt DC power source and output lead (booster and filter circuit terminal) connected to an associated magneto shall be energized and subjected to the following endurance test. With the magneto operating at 100 rpm, the coil shall be cycled for 250 hours at the rate of two minutes on time and four minutes off time. The coil shall subsequently be tested as specified in 4.6.1, at 77° ± 15°F.

4.6.3 Dielectric strength. To determine conformance to 3.3.3, the coil shall be subjected to a potential of 600 volts root mean square, 60 cycles per second applied for one minute between each output terminal and the case. The coil shall subsequently be tested as specified in 4.6.1, at 77° ± 15°F.

4.6.4 Shunting resistance. To determine conformance to 3.3.4, the coil shall be connected as specified in 4.6.1 with the addition of a 250,000 ohm, 20 kilovolt fixed resistor connected from magneto output to ground. The magneto shall be operated at a shaft speed of 100 rpm, with an input voltage of 28.5 ± 0.5 volts DC at an ambient air temperature of 77° ± 15°F. The output voltage shall be measured and recorded.

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4.7 Environmental.

4.7.1 Vibration. To detremine conformance to 3.4.1, the coil shall be mounted as in intended use. The coil shall then be vibrated for one hour in each direction of each of the three major axes as specified in method 201 of MIL-STD-202. Subsequently the coil shall be tested as specified in 4.6.1, at $77^{\circ} \pm 15^{\circ}F$.

4.7.2 Waterproofness. To determine conformance to 3.4.2, the coil with terminals uncovered shall be tested for waterproofness in accordance with MIL-E-13856 for type II, class 2 components.

4.7.3 Shock. To determine conformance to 3.4.3, the coil shall be mounted to simulate actual mounting in use and subjected to the shock test as specified in method 202 of MIL-STD-202 in such manner that an acceleration of 30 gravity units will be obtained. The test shall be repeated three times in each direction of each of the three major axes of the specimen. Subsequently the coil shall be tested as specified in 4.6.1, at $77^{\circ} \pm 15^{\circ}F$.

4.7.4 Fungus resistance. To determine conformance to 3.4.4, the coil shall be subjected to the fungus resistance test as specified in class 1, method B of MIL-F-13927, except that duration of the test shall be for 90 continuous days. Subsequently the coil shall be subjected to the tests specified in 4.6.1 and 4.6.3, at $77^{\circ} \pm 15^{\circ}F$.

4.7.5 Corrosion resistance. To determine conformance to 3.4.5, the coil shall be subjected to the salt spray test as specified in method 101 of MIL-STD-202 for a period of 200 hours. Subsequently the coil shall be tested as specified in 4.6.3 and 4.6.1, at $77^{\circ} \pm 15^{\circ}F$.

4.8 Inspection of preparation for delivery.

4.8.1 Materials and processes. The Government inspector shall, at unscheduled intervals, inspect all materials and processes involved in the preparation for delivery, to determine conformance to requirements of Section 5. Any evidence of deviation from specified requirements shall be cause for refusal to conduct further inspection until objective evidence has been provided by the contractor that corrective action has been taken.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, drying, preservation, packaging, packing and marking. Cleaning, drying, preservation, packaging, packing and marking shall be in accordance with the detailed preparation for delivery instructions provided by the procuring activity (see 6.2). When no detailed instructions are provided, cleaning, drying, preservation, packaging, packing and marking shall be in accordance with the applicable level of MIL-E-17555 (see 6.2).

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6. NOTES

6.1 Intended use. The coils covered by this specification are intended to provide additional starting current for the direct coupling type magneto of military vehicular internal combustion engines. The coil increases the magneto output at low cranking speeds, resulting in an intensified spark and easier starting.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Military part number.
(c) Other applicable standards or drawings if required (see 3.2).
(d) Treatment and painting if other than specified (see 3.5).
(e) Detailed preparation for delivery instructions (see 5.1).
(f) Selection of applicable level of packaging and packing of MIL-E-17555 required (see 5.1).
(g) Magnetos with which coil is intended to operate.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ, U.S. Army Tank-Automotive Command, ATTN: AMSTA-RSR, Warren, Michigan 48090 and information pertaining to qualification of products may be obtained from that activity.

Custodians:
Army - AT
Air Force - 82

Review activity:
DSA - CS

User activities:
Army - ME - EL
Navy - MC

Preparing activity:
Army - AT
Project No. 2920-0215

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8379

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PLATE NO. 15419 (BACK)

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